

Auteur: Sabawoon Enayat
Studierichting: Informatica
Oefeningenreeks 5D nr. 4.11

$$\text{Opgeve: } y_1 = x^3 + 1 \text{ en } y_2 = x^2 + x + 1$$

$$\Rightarrow y_1 = y_2 \Rightarrow y_2 - y_1 = 0 \Rightarrow (x^2 + 2x + 1) - (x^3 + 1) = 0$$

$$\Rightarrow -x^3 + x^2 + 2x = 0 \Rightarrow x(-x^2 + x + 2) = 0$$

$$D = 1 - 4 \cdot (-1) \cdot 2 = 9 \Rightarrow x_{1,2} = \frac{-1 \pm 3}{-2} \Rightarrow x_1 = -1, x_2 = 2$$

$$\Rightarrow x(x+1)(x-2) = 0 \Rightarrow x \in \{-1, 0, 2\}$$

$$\text{Stel } x = -\frac{1}{2} \Rightarrow y_1 = 1 - \frac{1}{8} = \frac{7}{8}, y_2 = 1 - 1 + \frac{1}{4} = \frac{1}{4} \Rightarrow y_1 > y_2$$

$$\text{Stel } x = 1 \Rightarrow y_1 = 1, y_2 = 1 + 2 + 1 = 4 \Rightarrow y_1 < y_2$$

$$\Rightarrow S = \int_{-1}^0 ((x^3 + 1) - (x^2 + 3x + 1)) dx + \int_0^2 ((x^2 + 2x + 1) - (x^3 + 1)) dx$$

$$= \int_{-1}^0 (x^3 - x^2 - 2x) dx + \int_0^2 (-x^3 + x^2 + 2x) dx$$

$$\Rightarrow S_1 = \int_{-1}^0 (x^3 - x^2 - 2x) dx$$

$$\Rightarrow \int (x^3 - x^2 - 2x) dx = \int x^3 dx - \int x^2 dx - 2 \int x dx = \frac{x^4}{4} - \frac{x^3}{3} - x^2 + c$$

$$\Rightarrow S_1 = \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_{-1}^0 = \left[0 - \left(\frac{1}{4} + \frac{1}{3} - 1 \right) \right] = 1 - \frac{1}{4} - \frac{1}{3} = \frac{12 - 3 - 4}{12}$$

$$\Rightarrow S_1 = \frac{5}{12}$$

$$\Rightarrow S_2 = \int_0^2 (-x^3 + x^2 + 2x) dx$$

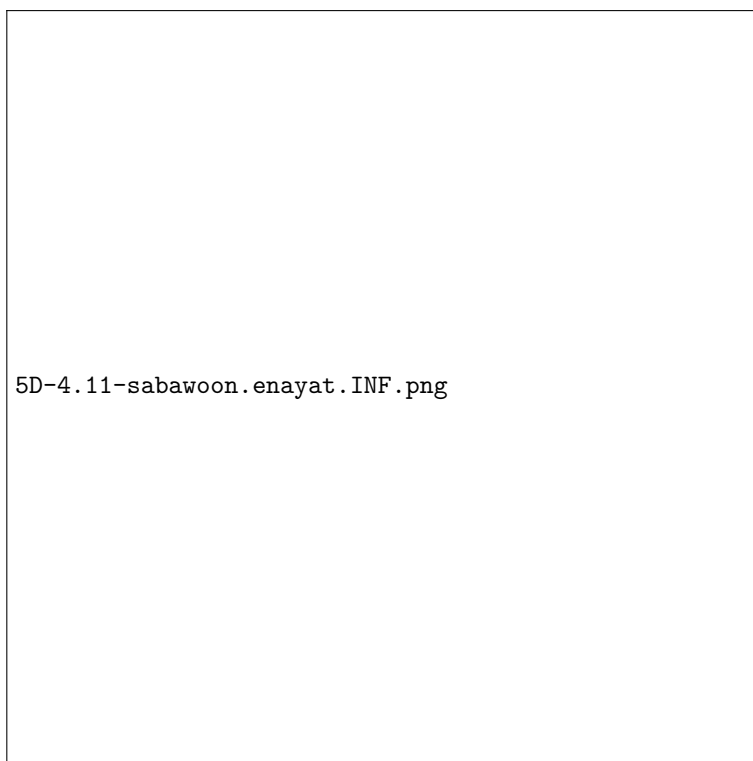
$$\Rightarrow \int (-x^3 + x^2 + 2x) dx = - \int x^3 dx + \int x^2 dx + 2 \int x dx$$

$$\Rightarrow S_2 = -\frac{x^4}{4} + \frac{x^3}{3} + x^2 + c$$

$$\Rightarrow S_2 = \left[-\frac{x^4}{4} + \frac{x^3}{3} + x^2 \right]_0^2 = \left[\left(-4 + \frac{8}{3} + 4 \right) - 0 \right] = \frac{8}{3}$$

$$\Rightarrow S = S_1 + S_2 = \frac{5}{12} + \frac{8}{3} = \frac{5 + 32}{12} = \frac{37}{12}$$

References



5D-4.11-sabawoon.enayat.INF.png